

TAMPA KNIGHT, FL, USA

WMO: 720374

Lat: 27.916N

Lon: 82.449W

Elev: 8

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 92825

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.5	44.7	25.0	19.0	48.1	29.7	23.7	51.4	18.9	65.7	17.4	64.3	7.1	40	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.4	94.9	78.8	92.9	78.4	91.2	78.0	82.5	88.7	81.8	87.9	81.2	87.1	6.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.3	162.9	84.5	80.9	160.3	84.3	79.4	152.6	83.8	46.3	87.7	45.4	89.5	44.7	86.5	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	16.0	13.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			31.1	82.6	4.3	3.3	28.0	85.0	25.5	87.0	23.1	88.8	20.0	91.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.7	60.8	64.6	68.7	75.3	80.0	83.0	83.7	84.1	82.8	76.4	69.0	67.6
	DBStd	9.60	8.19	6.99	7.15	4.36	3.83	2.87	2.75	2.47	2.68	5.47	6.69	7.06
	HDD50	20	17	3	0	0	0	0	0	0	0	0	0	0
	HDD65	416	175	86	50	1	0	0	0	0	0	7	38	59
	CDD50	9044	350	411	580	758	929	991	1046	1058	985	819	571	546
	CDD65	3968	44	74	165	310	464	541	581	593	535	362	159	140
	CDH74	37942	169	321	1041	2495	4552	5988	6523	6746	5648	2853	912	694
	CDH80	14550	8	26	222	774	1876	2546	2783	2842	2292	896	181	104
(14) Wind	WSAvg	6.2	6.2	6.0	6.8	6.9	6.6	5.6	4.8	5.3	6.2	6.9	6.8	6.3
(15) Precipitation	PrecAvg	52.1	2.6	2.6	3.2	2.6	2.9	8.1	8.1	8.7	7.4	2.4	1.6	2.3
	PrecMax	67.4	5.8	10.2	9.3	8.1	9.8	15.4	13.0	14.6	16.0	5.0	5.1	14.6
	PrecMin	36.6	0.4	0.3	0.4	0.2	0.2	2.7	3.9	5.1	2.3	0.2	0.2	0.0
	PrecStd	8.4	1.5	2.2	2.5	1.5	2.3	3.1	2.2	2.8	3.4	1.2	1.1	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.3	82.2	87.5	91.0	95.4	95.4	96.7	96.6	95.0	90.8	89.6	86.2
		MCWB	70.1	69.6	71.5	72.1	75.0	78.1	79.6	80.2	78.5	77.5	77.1	74.7
	2%	DB	78.9	80.8	84.3	88.5	91.5	93.3	93.4	93.5	92.6	89.7	84.2	82.1
		MCWB	68.7	68.9	69.6	71.4	73.2	78.0	79.4	80.4	78.4	76.8	72.9	72.4
	5%	DB	75.3	78.8	82.1	87.6	90.5	91.2	91.3	91.4	90.9	87.8	81.6	81.0
		MCWB	67.7	67.8	69.0	71.8	73.2	77.5	79.4	79.7	78.4	75.6	70.9	71.6
	10%	DB	72.7	75.1	80.8	84.3	89.5	90.3	90.4	90.5	90.1	85.6	79.3	78.7
		MCWB	66.1	65.9	68.7	70.4	73.2	77.4	79.0	79.4	78.1	74.0	69.1	70.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.1	74.4	75.8	78.2	80.0	83.5	83.3	83.4	82.7	80.9	78.4	76.9
		MCDB	78.2	79.8	82.5	85.5	86.1	89.9	89.9	90.3	88.2	85.6	84.3	82.2
	2%	WB	72.0	72.5	74.3	76.7	78.3	81.7	82.2	82.4	81.5	79.5	76.2	75.1
		MCDB	75.2	77.6	80.1	83.7	85.3	88.5	88.5	88.8	87.1	83.6	82.1	80.7
	5%	WB	69.9	70.5	72.6	75.1	77.4	80.9	81.4	81.9	80.9	78.3	73.9	73.7
		MCDB	73.0	74.6	77.9	82.4	84.9	87.3	87.4	88.0	86.4	83.4	79.1	78.4
	10%	WB	67.4	69.1	71.0	73.6	76.2	79.8	80.9	81.2	80.0	77.0	72.1	72.3
		MCDB	72.9	72.6	76.5	80.3	84.2	85.8	86.7	87.0	85.5	82.6	76.8	76.7
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	17.9	18.3	17.8	17.5	14.4	13.5	13.4	14.6	15.9	17.6	16.8
		MCDBR	18.5	18.3	19.1	18.9	19.0	15.9	14.8	14.7	15.2	16.1	17.2	16.6
		MCWBR	10.2	9.5	7.5	6.3	5.8	5.8	5.5	5.4	5.4	6.0	7.9	8.3
	5% WB	MCDBR	14.8	15.3	15.3	15.4	15.7	13.7	13.2	13.6	13.5	13.7	13.8	14.4
		MCWBR	10.8	10.4	8.4	8.0	5.9	6.1	5.9	5.7	5.6	5.9	7.7	8.2
(40) Clear-Sky Solar Irradiance	taub		0.342	0.355	0.368	0.391	0.425	0.450	0.474	0.455	0.422	0.390	0.354	0.342
	taud		2.520	2.493	2.443	2.343	2.307	2.316	2.252	2.321	2.382	2.454	2.499	2.547
	Ebn at Noon		286	292	293	287	276	267	261	266	272	276	279	281
	Edh at Noon		29	32	36	40	42	41	44	41	37	33	29	27
(44) All-Sky Solar Radiation	RadAvg		1124	1356	1712	2050	2168	1953	1853	1786	1627	1501	1225	1039
	RadStd		80	86	138	100	121	158	114	112	103	93	62	85

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-7	-85	+295	+209

Nomenclature: See separate page